Confirm that the retarded potentials satisfy the Lorenz gauge condition. [Hint: First show that

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath} (\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right)$$

where $\boldsymbol{\nabla}$ denotes derivatives with respect to $\mathbf{r}$, and $\boldsymbol{\nabla}'$ denotes derivatives with respect to $\mathbf{r}'$. Next, noting that $\mathbf{J}(\mathbf{r}', t - \imath/c)$ depends on $\mathbf{r}'$ both explicitly and through $\imath$, whereas it depends only on $\mathbf{r}$ through $\imath$, confirm that

$$\boldsymbol{\nabla} \cdot \mathbf{J} = -\frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath), \boldsymbol{\nabla}' \cdot \mathbf{J} = -\dot{\rho} - \frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath)$$

Use this to calculate the divergence of $\mathbf{A}$ (Eq. 10.26.)]

Solution:

We want to show

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

where

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{\imath} \, \mathrm{d}\tau'$$

Now, taking the divergence of $\mathbf{A}$ gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}(\mathbf{r}', t_r)}{\imath} \right) \mathrm{d}\tau'$$

From now on, $\mathbf{J}(\mathbf{r}', t_r) = \mathbf{J}$. Using a divergence identity gives

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla} \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath} \right)$$

and

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla}' \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\boldsymbol{\nabla}' \frac{1}{\imath} \right)$$

Since $\frac{1}{\imath} = \frac{1}{|\mathbf{r} - \mathbf{r}'|}$, we have

$$\boldsymbol{\nabla}' \frac{1}{\imath} = -\boldsymbol{\nabla} \frac{1}{\imath}$$

which means we can rewrite $\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right)$ as

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla}' \cdot \mathbf{J}) - \mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath} \right)$$

$$\downarrow$$

$$\mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right)$$

Putting this expression into $\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right)$ gives

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$$

which is what the hint wanted us to show. Putting this into the expression for $\boldsymbol{\nabla} \cdot \mathbf{A}$ gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \left[\frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)\right] \mathrm{d}\tau' \tag{1}$$

According to the Lorenz gauge condition,

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t} = -\frac{\mu_0}{4\pi} \int \frac{\dot{\rho}}{\imath} \, \mathrm{d}\tau'$$

With this information, we now evaluate the three terms in Equation 1

Starting with the first term, and using the chain rule on $\mathbf{J}$. Since $\mathbf{J}$ depends on $\mathbf{r}$ only through t_r,

$$\begin{gathered} \boldsymbol{\nabla} \cdot \mathbf{J} = \dot{\mathbf{J}} \cdot \boldsymbol{\nabla} t_r = -\frac{1}{c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath) \\ \downarrow \\ \frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) = -\frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath) \end{gathered} \tag{2}$$

The next term is the divergence of $\mathbf{J}$ with respect to $\mathbf{r}'$. $\mathbf{J}$ depends on $\mathbf{r}'$ both explicitly and through t_r, so

$$\begin{gathered} \boldsymbol{\nabla}' \cdot \mathbf{J} = (\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r} + \dot{\mathbf{J}} \cdot \boldsymbol{\nabla}' t_r = -\dot{\rho} - \frac{1}{c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath) \\ \downarrow \\ \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) = -\frac{\dot{\rho}}{\imath} - \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath) = -\frac{\dot{\rho}}{\imath} + \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath) \end{gathered} \tag{3}$$

where $(\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r}$ is the divergence taken at fixed t_r, which equals $-\dot{\rho}$ by the continuity equation.

The last term integrates to 0 which I will explain later.

Adding Equation 2 and Equation 3, the only term left is

$$-\frac{\dot{\rho}}{\imath}$$

which gives

$$\boldsymbol{\nabla}\cdot\mathbf{A}=\frac{\mu_0}{4\pi}\int\left(-\frac{\dot{\rho}}{\imath}\right)\mathrm{d}\tau'=-\frac{\mu_0\epsilon_0}{4\pi\epsilon_0}\int\frac{\dot{\rho}}{\imath}\,\mathrm{d}\tau'=$$

$$=-\mu_0\epsilon_0\frac{\partial}{\partial t}\left[\frac{1}{4\pi\epsilon_0}\int\frac{\rho}{\imath}\,\mathrm{d}\tau'\right]=-\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

$$\downarrow$$

$$\boxed{\boldsymbol{\nabla}\cdot\mathbf{A}=-\mu_0\epsilon_0\frac{\partial V}{\partial t}}$$

□
